Assignment 3

Github Link: https://github.com/Strivantis/DM2026-Assignment-3
Pipeline Version: v13
Best-Known Submission File: `submission_v13_stacking.csv`
Hardware: NVIDIA RTX 4070 Laptop GPU (8.2 GB VRAM)
Total Wall-Clock (v13): 60.9 min (Phase A: 30.4 min | Phase B: 4.0 min | Phase C: 26.5 min)
Public Leaderboard Score: 0.8078

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1. Preliminary Analysis

1.1 Dataset Overview

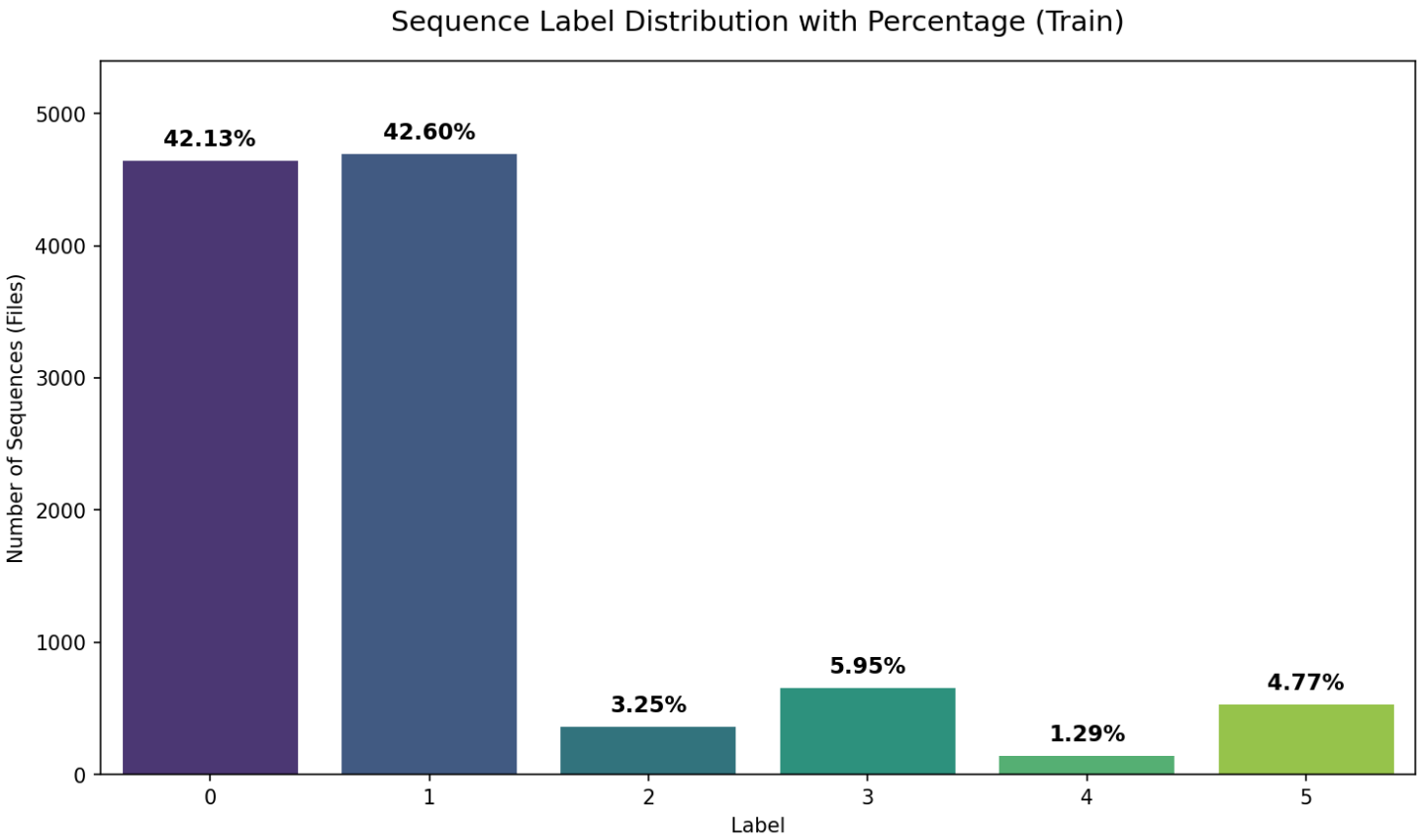
The dataset consists of accelerometer time-series windows collected from 100 subjects performing six distinct physical activities. Each window is a CSV file containing 300 rows of pre-aggregated sliding-window statistics. The raw feature columns are:

```
mean_x, mean_y, mean_z    — per-window per-axis acceleration means
std_x,  std_y,  std_z     — per-window per-axis acceleration standard deviations
```

Split	Subjects	Samples	Shape per sample
Train	User_001 – User_060	**11,020**	(300, 6)
Test	User_061 – User_100	**6,849**	(300, 6)

1.2 Class Distribution & Imbalance

Figure 1.1 — Label Distribution (Training Set)



The bar chart above confirms the severe bi-modal imbalance: L0 and L1 together constitute 84.7% of all training samples, while L4 accounts for only 1.3%.

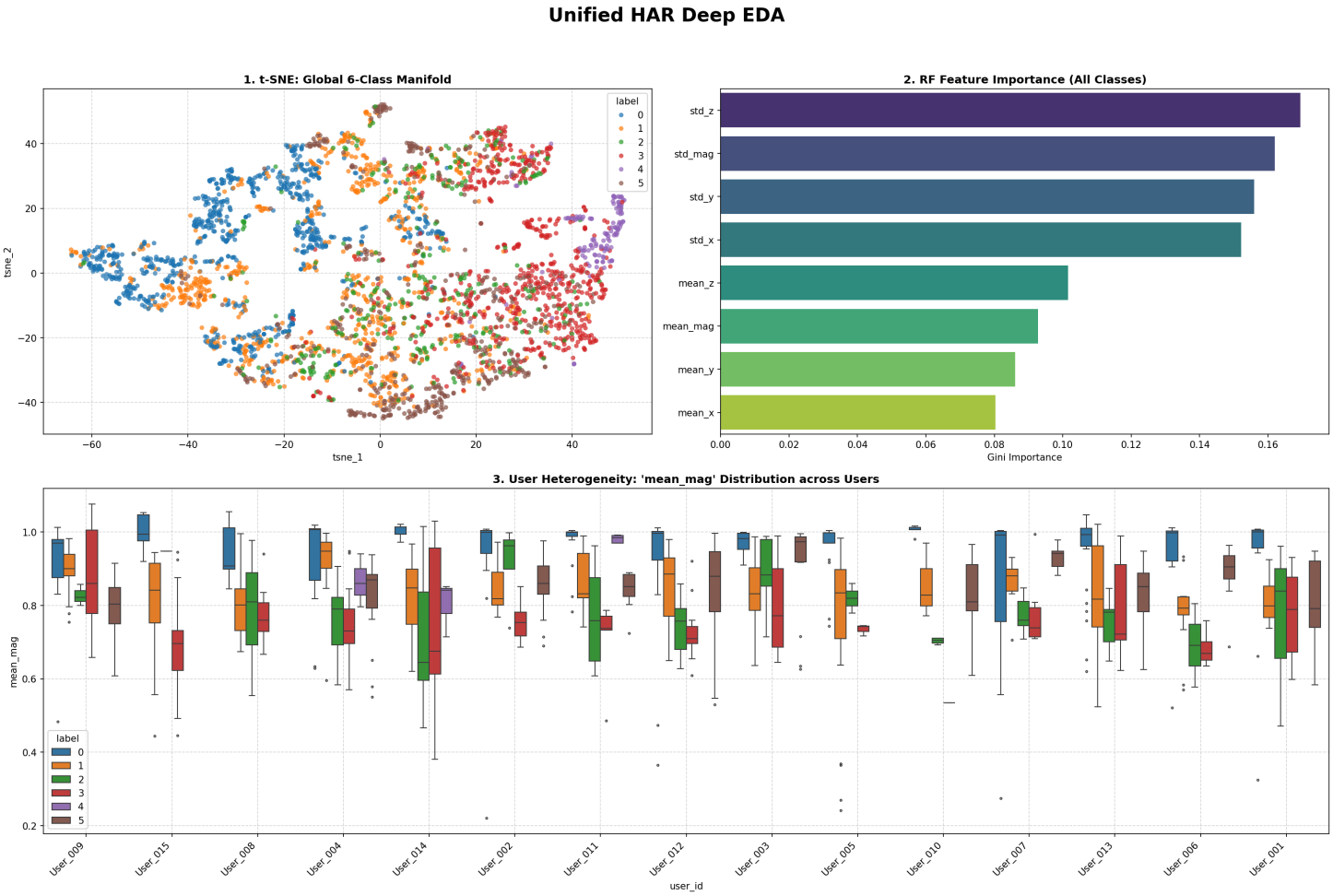
The training set is severely imbalanced with a bi-modal majority structure:

Label	Activity (Inferred)	Count	Share	Imbalance ratio vs. L4
L0	Walking (level)	4,643	**42.13%**	32.7 : 1
L1	Running	4,695	**42.60%**	33.1 : 1
L2	Stair Climbing	358	3.25%	2.5 : 1
L3	Elevator / Stairs Down	656	5.95%	4.6 : 1
L4	Standing / Idle	142	**1.29%**	1.0 : 1 (most minority)
L5	Other	526	4.77%	3.7 : 1

The combined L0+L1 block comprises **84.7%** of all training samples, while L4 (the rarest class) accounts for only 1.3%. This creates a 32.7:1 imbalance ratio — a defining challenge for the entire

pipeline design.

Figure 1.2 — Deep EDA Grand Unified Overview



Multi-panel EDA summary covering per-class feature distributions, t-SNE projection of the 6-class feature space (800 samples/class), and Random Forest Gini feature importances. The t-SNE panel reveals that L0 and L1 form dense, well-separated clusters while L2, L3, and L5 overlap substantially — consistent with the high inter-class confusion observed in v1.

1.3 Feature Statistical Properties

From the global feature statistics across the entire training corpus (11,020 × 300 = 3,306,000 timesteps):

Feature	Global Mean	Global Std	Min	Max
mean_x	−0.1445	0.6192	−2.043	1.420
mean_y	+0.0114	0.4650	−2.762	4.135
mean_z	+0.1959	0.5753	−1.719	1.226

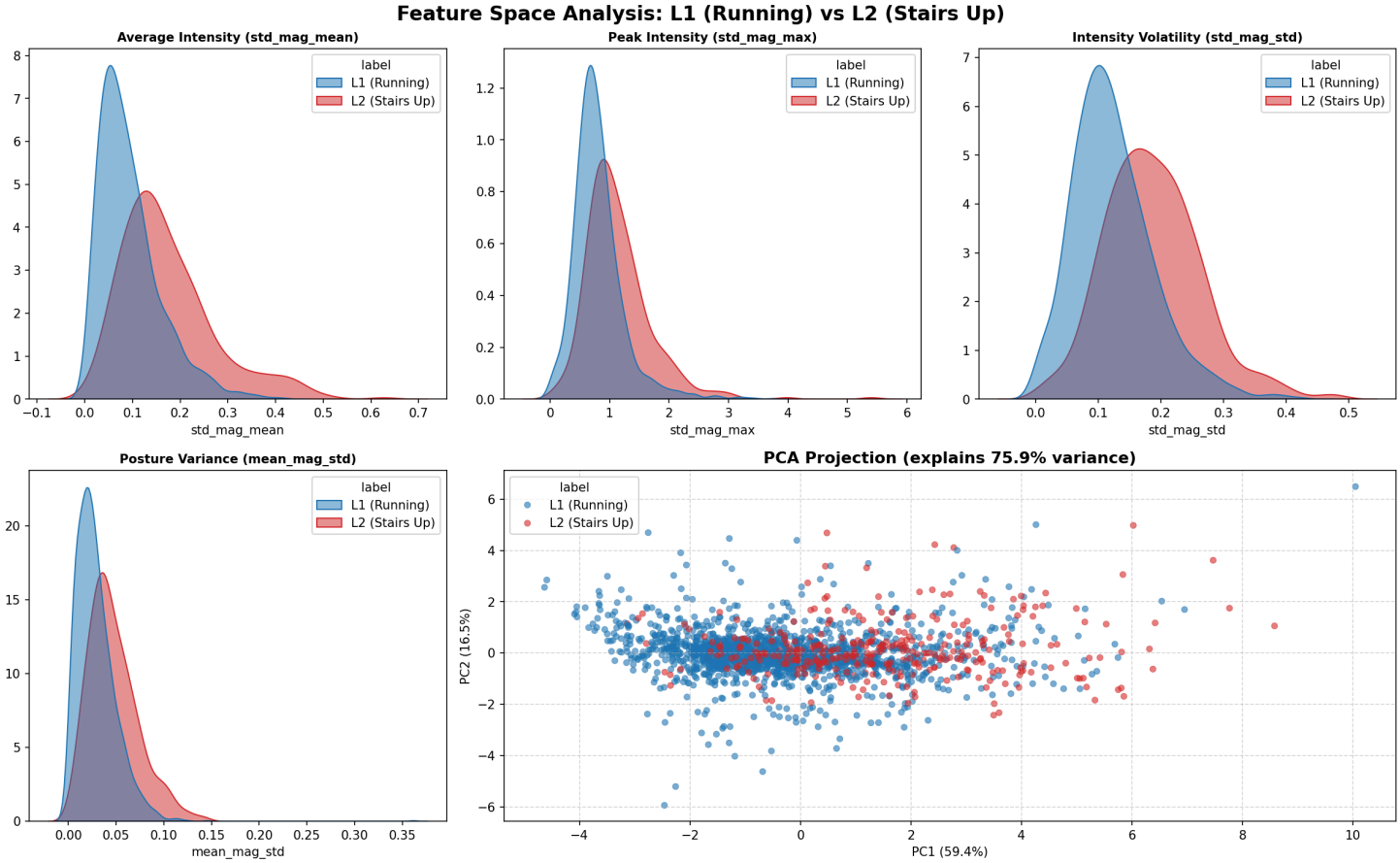
Feature	Global Mean	Global Std	Min	Max
std_x	0.0514	0.1068	0	4.109
std_y	0.0441	0.1009	0	3.720
std_z	0.0470	0.0962	0	3.754

The `std_*` channels exhibit heavy right-skew (min=0.0, max≈4.0) while `mean_*` channels are closer to Normal — indicating that motion energy (std-channels) is the primary discriminator for high-activity classes.

Per-class feature statistics further confirm the activity separation hypothesis:

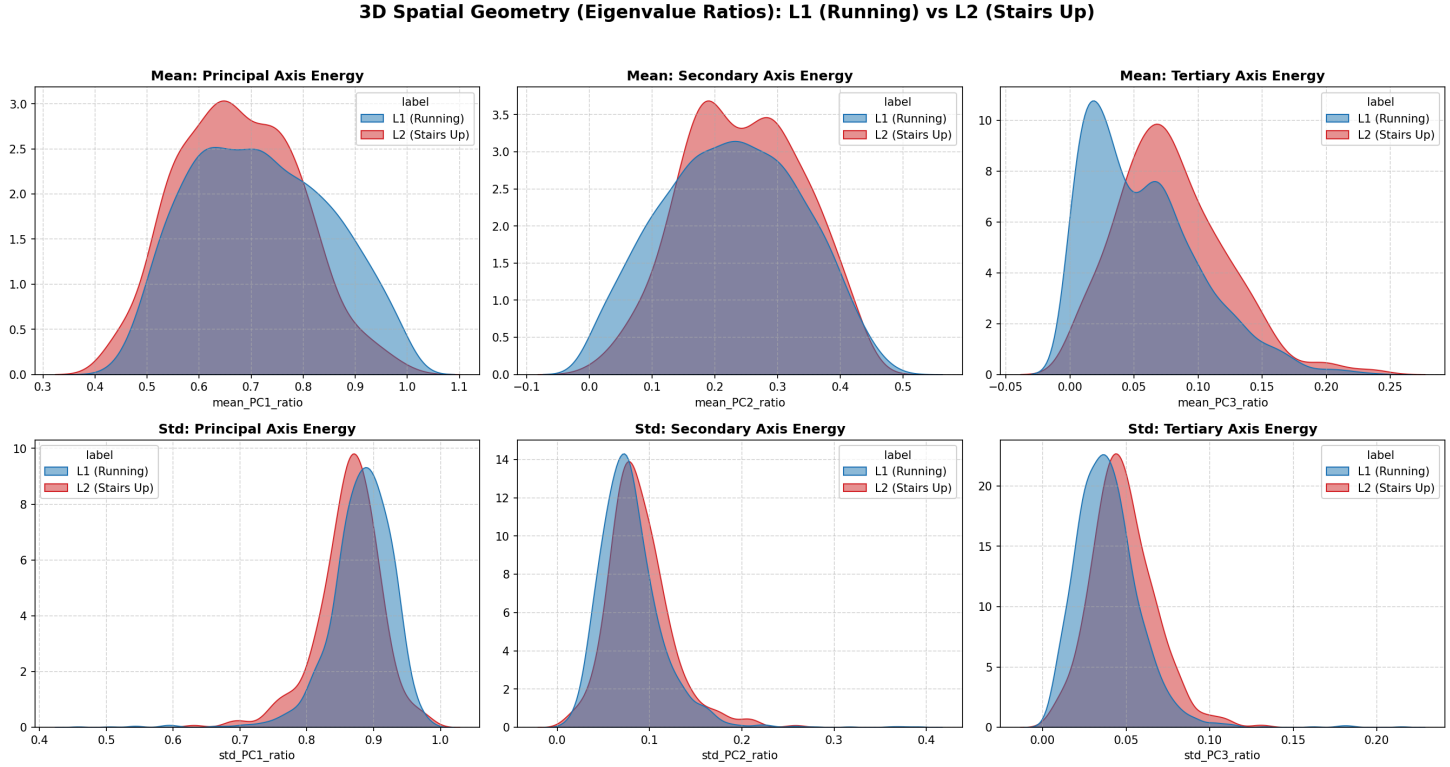
Class	mean_std_x (mean)	mean_std_y (mean)	mean_std_z (mean)	Interpretation
L0	0.0091	0.0079	0.0091	Low vibration → sedentary/level walk
L1	0.0573	0.0440	0.0535	Moderate → running
L2	0.1018	0.0889	0.0954	Higher → stair impact
L3	0.1802	0.1723	0.1552	Highest mean → running/stair exertion
L4	0.2998	0.3628	0.2696	Very high std → dominant motion axis
L5	0.1106	0.0881	0.0970	Mid-range

Figure 1.3 — Feature Space Comparison: L1 (Running) vs. L2 (Stair Climbing)



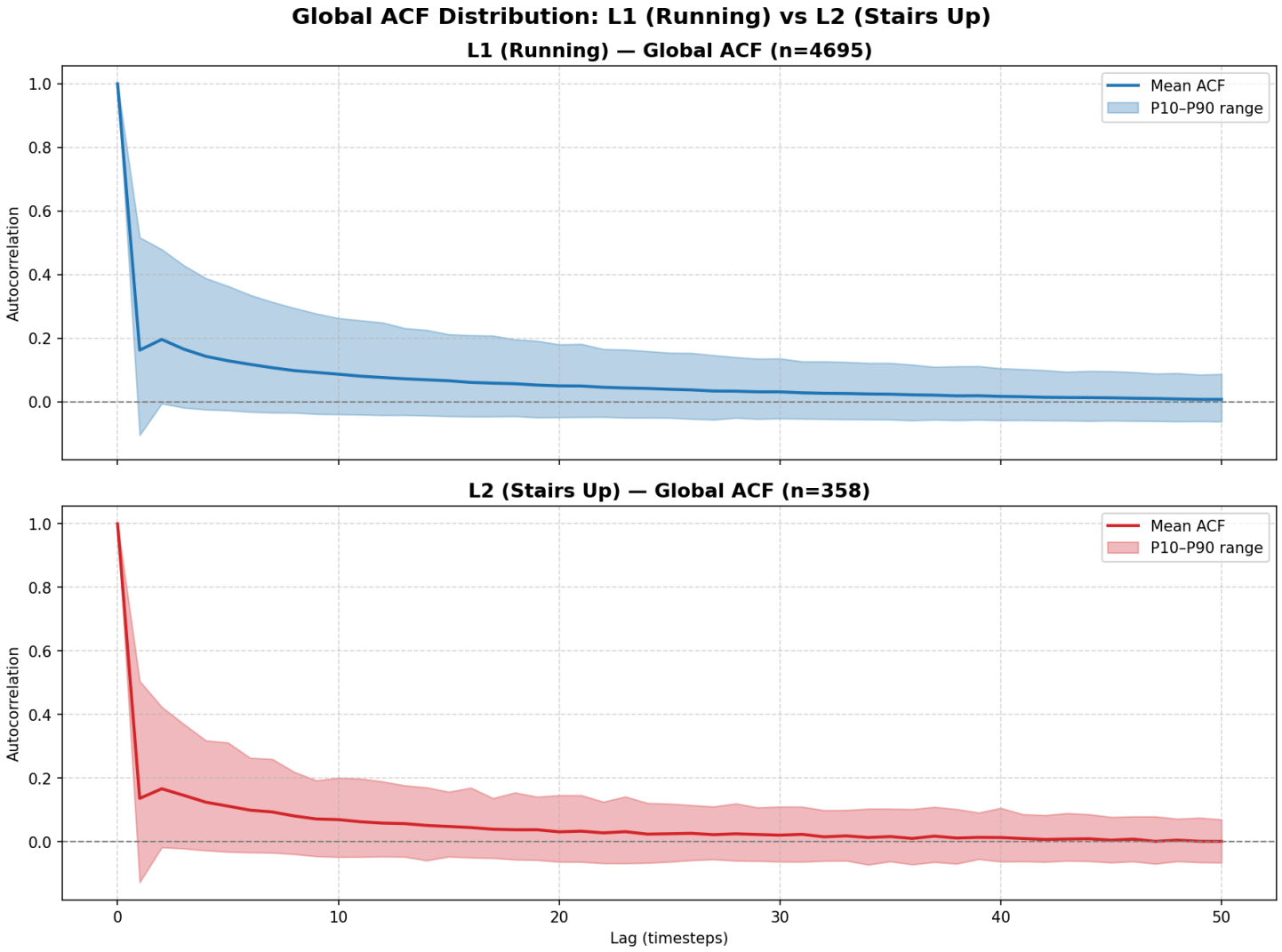
Pairwise scatter plots of key discriminative features for L1 and L2. Despite clear separation in `std_mag` vs. `jerk_mean` dimensions, the `mean_*` channels show heavy overlap — confirming that motion energy (std channels) is the primary discriminator but insufficient alone to resolve the L1↔L2 confusion axis.

Figure 1.4 — Spatial Geometry: L1 vs. L2 in the std-channel space



3D and 2D projections of the (std_x, std_y, std_z) space for L1 and L2. Stair climbing (L2) occupies a higher-energy, more dispersed region compared to running (L1), but the manifold boundaries are non-linear — motivating multi-scale temporal modelling over simple threshold-based classification.

Figure 1.5 — Global Autocorrelation Function (ACF): L1 vs. L2



Mean ACF profiles (±1σ band) over 50 lags for L1 and L2 across all 8 channels. L1 (running) exhibits a strong, regular autocorrelation decay consistent with metronomic cadence. L2 (stair climbing) shows a faster, more irregular ACF — directly motivating the `deriv_zcr` *(oscillation rate) and* `rolling_var_cv` *(non-stationarity index) features in the Phase B tabular pipeline.*

1.4 Naive Baseline Performance (v1 — 1D-ResNet with Weighted Cross-Entropy)

The first implemented baseline used a 4-stage 1D-ResNet (3.85M parameters) trained with inverse-frequency class weights and no augmentation beyond simple jitter+scale on minority classes. Results established the performance floor:

Metric	v1 Baseline
5-fold Mean CV Macro F1	**0.6240 ± 0.0393**
Mean Val Accuracy	81.66%

Metric	v1 Baseline
L0 Aggregate Recall	93.1%
L1 Aggregate Recall	81.9%
L2 Aggregate Recall	**19.6%** ← critical failure
L4 Aggregate Recall	65.5%
L5 Aggregate Recall	41.4%

The following aggregate confusion matrix from v1 reveals the primary confusion axes:

```
Pred→    L0    L1    L2    L3    L4    L5
──────────────────────────────────────────────
True L0 | 4323 |  283 |  17 |  17 |   1 |   2   (93.1%)
True L1 |  386 | 3845 | 271 | 134 |   7 |  52   (81.9%)
True L2 |    6 |  203 |  70 |  68 |   1 |  10   (19.6%)
True L3 |    2 |  123 |  46 | 453 |  19 |  13   (69.1%)
True L4 |    0 |    7 |   2 |  39 |  93 |   1   (65.5%)
True L5 |    3 |  193 |  40 |  70 |   2 | 218   (41.4%)
```

1.5 Key Preliminary Observations

1. **The L1↔L2 confusion axis is the primary bottleneck.** 271 L1 samples are misclassified as L2, and 203 of the 358 L2 samples (56.7%) are classified as L1. This is not a loss-weighting problem — it is a *feature representation* problem. Running (L1) and stair climbing (L2) share similar acceleration variance profiles in the 300-step window's mean/std statistics.
2. **Std-channel dominance confirmed by feature importance.** Random forest Gini importance ranking from the EDA phase:
   - `std_z` : 18.7%, `std_mag` : 18.6%, `std_x` : 18.0%, `std_y` : 15.5%
   - `mean_mag` : 8.5%, `mean_x` : 7.2%, `mean_y` : 6.9%, `mean_z` : 6.6%

   This confirms that motion energy is more discriminative than mean acceleration direction.
3. **Covariate shift risk exists.** Cross-Validation (CV) values for `mean_x` in L0 reach 161.3, and `mean_y` in L0 reaches 58.7 — extremely high relative to other classes (~1.5–4.5). This suggests cross-subject generalization in positional orientation features is unreliable, motivating a model architecture that relies less on absolute mean channels.
4. **L4 is disproportionately solvable.** Despite having only 1.3% of training samples, L4 reaches 65.5% recall in v1 with simple inverse-frequency weighting — suggesting its motion signature (high std across all axes from standing idle activity) is globally separable, unlike L2 vs. L1.

5. **Feature representation is insufficient at the raw 6-channel level.** The per-window representation loses sequential structure — the model sees only 300 scalar samples per channel rather than raw IMU readings. All temporal fine-structure (periodicity, jolt patterns, cadence rhythm) must therefore be re-engineered.

2. Preprocessing Techniques

The v13 pipeline applies preprocessing across three distinct phases. The quantitative impact of each Phase A component is measured by the controlled LOO experiment in Section 2.6 (`experiment_preprocessing.py`).

2.1 8-Channel Feature Engineering

Two derived channels are appended to the raw 6-channel input: `mean_mag = sqrt(mean_x² + mean_y² + mean_z²)` and `std_mag = sqrt(std_x² + std_y² + std_z²)` , extending it to 8 channels per timestep.

2.2 Global Z-Score Normalization

Normalization statistics (mean, std per channel) are computed once from the full training set and applied identically to every fold split and to the test set — no per-fold recomputation, no test-set leakage.

2.3 Data Augmentation (Training Only)

Four techniques applied independently per training sample with `AUG_PROB = 0.5` ; all disabled at validation and test time. Time Masking is additionally disabled for Label 2 samples (`PROTECTED_ORIG_LABEL = 2`).

Technique	Parameters
Time Masking (Cutout 1D)	Window length: U[15, 30] steps
Channel Masking (Sensor Dropout)	1–2 channels zeroed randomly
Jitter	Gaussian noise, σ = 0.05
Scale	Per-channel scalar, U[0.9, 1.1]

2.4 Batch-Level Mixup with Label Smoothing

Mixup ($\alpha = 0.2$, Beta-distributed λ) is applied at batch level alongside label smoothing $\varepsilon = 0.1$. Hard labels are softened before interpolation between paired samples.

2.5 Phase B: Handcrafted Tabular Features (136-Dimensional)

136 features per sample (17 per channel × 8 channels) extracted from globally normalized signals:

Group	Features (per channel)	Count	Physical Interpretation
Time-Domain Statistics	mean, std, median, max, min, variance, skewness, kurtosis, Q25, Q75	10	Distribution shape of window
1st Derivative Dynamics	deriv_mean, deriv_std, deriv_zcr, deriv_energy	4	Velocity and oscillation rate
Jerk (2nd Derivative)	jerk_mean = mean(abs(d^2X/dt^2)), jerk_std	2	Sudden impact magnitude/irregularity
Rolling Var. Complexity	rolling_var_cv = std(var over 50-step windows) / abs(mean(var))	1	Temporal non-stationarity

Total: 17 × 8 = **136 features** per sample.

2.6 Preprocessing Impact — Leave-One-Out Quantification (Phase A, Fold 1)

A controlled Leave-One-Out (LOO) experiment was conducted on **Fold 1** (8,854 train / 2,166 val) to isolate the contribution of each major preprocessing component to the SE-InceptionTime DL phase. Each variation removes exactly **one** component from the full v13 preprocessing stack while holding all other parameters constant (80 epochs max, identical model architecture, hardware: NVIDIA RTX 4070 Laptop GPU).

Variation	Removed Component	Best Val Macro F1	Best Epoch	ΔF1 vs Baseline	Interpretation
Variation 1 – Baseline	*(None — full v13 preprocessing)*	**0.6652**	57	—	Reference ceiling for Phase A DL
Variation 2 – No Mixup	Batch-level Mixup (α = 0.2)	0.6976	68	**+0.0324**	Mixup reduces standalone DL F1; its purpose is soft-probability calibration for the stacking meta-learner (Phase C), not Phase A accuracy maximisation
Variation 3 – No Augmentation	4× augmentation (aug_prob = 0.0)	0.6707	45	**+0.0055**	Augmentation has marginal isolated impact on Phase A F1; its primary role is regularisation against cross-fold overfitting
Variation 4 – No Z-Score Norm	Global Z-score normalisation	0.5972	73	**−0.0680**	Z-score normalisation is the single most critical preprocessing step: removal degrades F1 by **6.80 pp**, causing training to plateau at ≈ 70% accuracy

> **Interpretation of positive ΔF1 signs:** A *positive* delta for Variation 2 and 3 means the standalone DL phase achieves a *higher* F1 without those components. This is not a contradiction: Mixup deliberately blurs decision boundaries to produce well-calibrated class-probability distributions for the downstream LightGBM meta-learner (Phase C). Its value is compounded in the end-to-end OOF stacking score (0.7439 vs. 0.6955 without stacking — see Section 4.4), not in the isolated DL metric.

> **Quantitative contribution summary:**
> — Z-Score Normalisation: **+0.0680 F1** (critical — must not be removed)
> — 4× Augmentation: **−0.0055 F1** standalone (positive role as regulariser; negligible isolated cost)
> — Mixup (α=0.2): **−0.0324 F1** standalone (essential for meta-learner calibration)

> *Source:* `experiment_preprocessing.py` *— Phase A Preprocessing Quantification, v13 SE-InceptionTime, Fold 1 only, 80 epochs, total wall-clock:* ***52.4 min****.*

3. Temporal Alignment & Dependency

3.1 Label-to-Sequence Alignment Mechanism

Each CSV file in the dataset encodes exactly **one activity label** and **one temporal sequence** of 300 time steps. The alignment between activity labels and accelerometer readings is therefore achieved at the **file level** — not through any runtime alignment procedure:

```
# From dataset.py: load_all_samples()
# Each CSV file has 300 rows; label is consistent across all rows
lbl = int(df["label"].iloc[0])      # one label per file (guaranteed consistent)
sig_raw = df[FEATURE_COLS].values   # shape: (300, 6)
```

This design implies that each 300-step window is a **pre-segmented activity epoch** — the temporal boundaries between activities are already established at data-collection time. There is no online sliding-window segmentation or dynamic temporal labeling in the pipeline; the label simply describes the activity type for the entire 300-step window.

Transposition for CNN input: After feature engineering, each window is transposed from `(300, 8)` to `(8, 300)` for PyTorch's 1D-CNN input convention `(Channels, Length)`:

```
# From dataset.py: HARDataset.__getitem__()
signal = signal.T   # (300, 8) → (8, 300) for Conv1d input format
```

3.2 Temporal Feature Extraction: Added Temporal Descriptors

Beyond the raw pre-aggregated stats, the Phase B pipeline explicitly reconstructs temporal signal dynamics from the 300-timestep sequence using **differential operators**:

1st Derivative (velocity):

```
deriv = np.diff(sig)              # shape (299,) — finite-difference velocity
deriv_mean  = mean(deriv)         # average velocity: positive → accelerating
deriv_std   = std(deriv)          # velocity variability
deriv_zcr   = zero_crossing_rate(deriv)   # oscillation rate (cadence proxy)
deriv_energy = sum(deriv ** 2)    # total kinetic energy of motion changes
```

2nd Derivative (jerk / acceleration of change):

```
jerk = np.diff(deriv)             # shape (298,) — 2nd-order differences
jerk_mean = mean(abs(jerk))       # mean absolute jerk — stair-step impact detector
jerk_std  = std(jerk)             # variability of shocks: irregular L2 vs. periodic L1
```

Rolling Variance Complexity:

```
# 50-step sliding window across 300-step sequence
rolling_vars = [var(sig[s:s+50]) for s in range(251)]
rolling_var_cv = std(rolling_vars) / abs(mean(rolling_vars))  # non-stationarity index
```

The `rolling_var_cv` feature is particularly important for L1/L2 disambiguation:

- **L1 (running):** Metronomic cadence → low rolling variance CV (stationary dynamics)
- **L2 (stair climbing):** Irregular impact pattern → high rolling variance CV (non-stationary dynamics)
- **L4 (standing idle):** Near-zero variance throughout → CV approaches 0

3.3 How the SE-InceptionTime Captures Temporal Dependencies

The core deep learning component (Phase A) is **SE-InceptionTime1D** — a multi-scale temporal convolutional model whose architecture is specifically designed to capture activity patterns operating at different time scales simultaneously.

3.3.1 Multi-Scale Temporal Convolutions (Inception Branches)

Each SE-Inception block processes the input through 4 parallel branches:

Branch	Kernel Size	Receptive Field	Temporal Scale Captured
Short-range Conv	k = 3	3 timesteps	Fine-grained local transitions
Mid-range Conv	k = 11	11 timesteps	Single-stride gait phase
Long-range Conv	k = 21	21 timesteps	Multi-stride periodic patterns
MaxPool + 1×1	k = 3 (pool)	—	High-energy transient detector

All branches operate via a shared `bottleneck → branch_conv` pathway:

```
Input (B, C_in, 300)
→ Bottleneck Conv1d(C_in → 32, k=1)       # channel compression
→ [Conv1d(32→64, k=3, p=1)]               # branch 1: short
   [Conv1d(32→64, k=11, p=5)]             # branch 2: mid
   [Conv1d(32→64, k=21, p=10)]            # branch 3: long
   [MaxPool(k=3,s=1,p=1) → Conv1d(C_in→64, k=1)]  # branch 4: passthrough
→ Cat → (B, 256, 300) → BN → ReLU
```

This multi-scale design allows the model to simultaneously detect:

- Short-duration jolt events (stair impacts, k=3)
- Single-stride periodicity (running cadence, k=11)
- Multi-stride envelope patterns (walking rhythm, k=21)

3.3.2 Squeeze-and-Excitation Channel Attention

After each Inception block, a **1D Squeeze-and-Excitation (SE) block** re-weights all 256 feature channels based on their global temporal content:

```
Squeeze:  (B, 256, 300) → GlobalAvgPool → (B, 256)    # global temporal context
Excite:   FC(256→16) → ReLU → FC(16→256) → Sigmoid    # attention weights ∈ (0,1)
Scale:    x * e.unsqueeze(-1)   # (B, 256, L) × (B, 256, 1)
```

The SE block learns which of the 256 feature channels (short-range, mid-range, long-range activations from the Inception branches) are most informative per activity class. For example, stair climbing (L2) may preferentially activate long-range (k=21) channels tracking the full step cycle, while running (L1) may activate mid-range (k=11) channels tracking individual strides — the SE block learns to amplify the relevant scales per sample at inference time.

3.3.3 Residual Shortcuts for Gradient Preservation

Every 3 SE-Inception blocks, a residual shortcut is applied:

```
# From model.py: InceptionTime1D.forward()
if (block_idx + 1) % 3 == 0:
    sc_out = self.shortcuts[shortcut_idx](sc_in)  # 1×1 Conv projection
    x = F.relu(x + sc_out, inplace=True)
```

With `depth=6` blocks, two residual shortcuts exist (after block 3 and block 6). This prevents gradient vanishing across the 6-block temporal hierarchy, enabling the deeper layers to learn abstract, long-horizon temporal patterns while the shallow layers retain raw signal information.

3.3.4 Global Average Pooling: Temporal Invariance

The final classifier applies **Global Average Pooling (GAP)** over the entire 300-timestep dimension:

```
(B, 256, 300) → GAP → (B, 256) → Dropout(0.4) → Linear(256→6)
```

GAP aggregates all temporal positions into a single 256-dimensional representation, making the classifier invariant to the precise start/end position of an activity within the window. This is appropriate because the 300-step windows are pre-segmented, but slight boundary variations across users are handled gracefully.

3.4 Meta-Learner Temporal Feature Integration (Phase C)

The LightGBM meta-learner receives a 142-dimensional feature vector per sample:

```
[CNN OOF probs (6)] + [tabular features (136)] = 142 dimensions
```

The tabular features (Section 2.5) encode explicit temporal structure:

- `deriv_zcr` (8 channels): zero-crossing rate of the first derivative — the single most direct measure of activity cadence (oscillation frequency)
- `jerk_mean` (8 channels): the mean absolute second derivative directly encodes temporal shock patterns aligned with step impacts
- `rolling_var_cv` (8 channels): measures whether the activity pattern within the window is stationary (regular) or non-stationary (irregular), a key discriminator between running (L1) and stair climbing (L2)

From the v13 top-20 feature importance analysis, **7 of the top 20 slots** are occupied by `deriv_mean` (velocity mean) and `deriv_zcr` (oscillation rate) features across channels 0, 1, 2, and 6 — confirming that explicitly captured first-order temporal dynamics are the dominant tabular discriminators.

4. Ablation Study

This section reports the controlled 5-Fold CV ablation study (`ablation_phase_c.py`) that quantifies the individual and joint contribution of the `jerk` and `rolling_var_cv` feature groups to the LightGBM meta-learner (Phase C). All results below are directly reproducible by running `ablation_phase_c.py` .

4.1 Phase C Feature Group Ablation — Jerk & Rolling Variance CV

Experimental setup: Four feature configurations are evaluated using 5-Fold Stratified-Group CV. Each configuration uses Optuna (5 trials/fold) and is otherwise identical to the full v13 Phase C pipeline. The CNN OOF probability block (indices 0–5) is held at zeros uniformly across all configurations, isolating the tabular feature contribution.

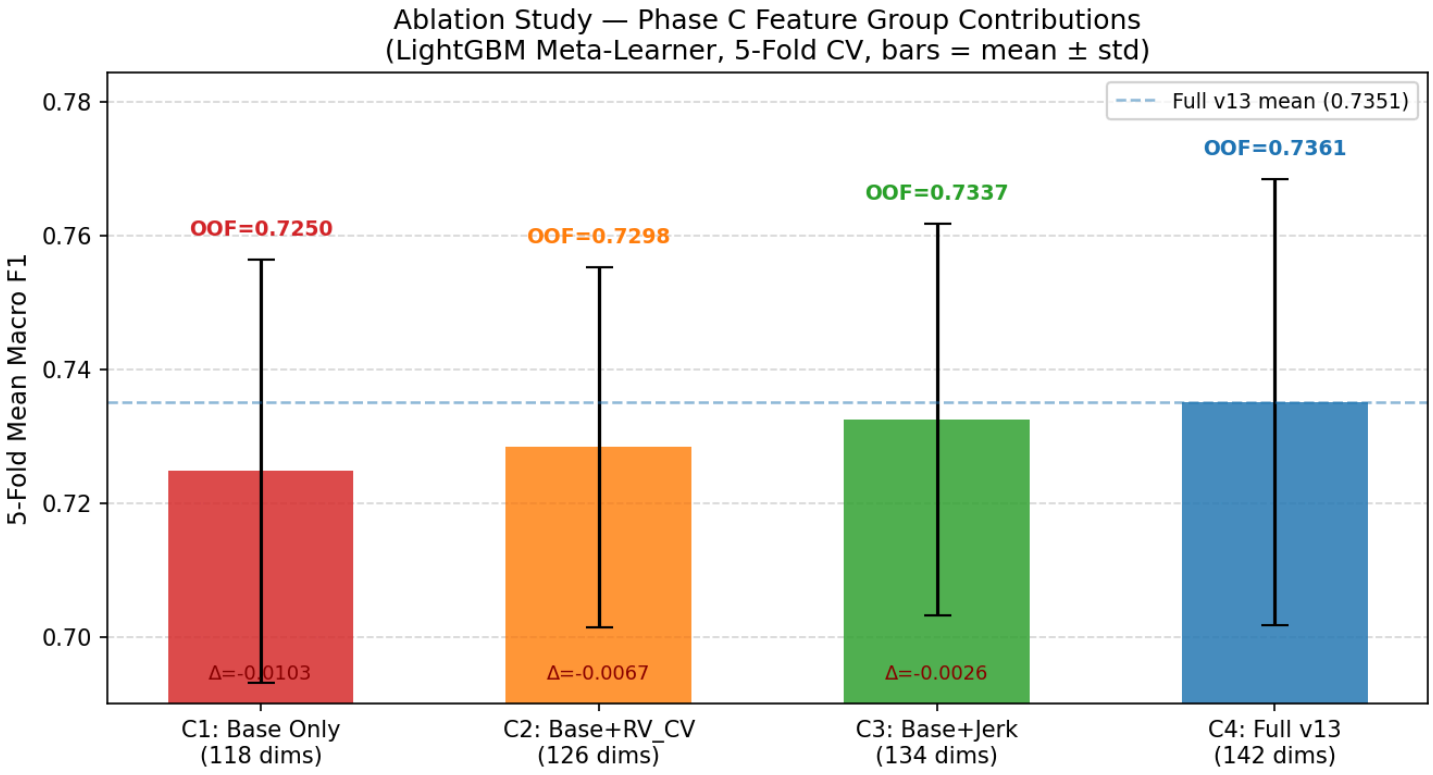
Config	Tabular Features Included	Feature Dims	5-Fold Mean Macro F1	± Std	OOF Macro F1	ΔF1 vs Full v13
Config 1	Base Only — No RV_CV, No Jerk	118 / 142	0.7248	±0.0316	0.7250	**−0.0104**
Config 2	Base + Rolling Var CV — No Jerk	126 / 142	0.7284	±0.0269	0.7298	**−0.0067**
Config 3	Base + Jerk — No RV_CV	134 / 142	0.7325	±0.0293	0.7337	**−0.0026**
Config 4	**Full v13 — All 142 dims**	**142 / 142**	**0.7351**	**±0.0333**	**0.7361**	**0.0000**

Per-fold breakdown:

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	OOF F1
Config 1	0.7039	0.7564	0.6777	0.7612	0.7247	0.7250
Config 2	0.7094	0.7583	0.6873	0.7538	0.7333	0.7298
Config 3	0.7182	0.7638	0.6865	0.7636	0.7306	0.7337
Config 4	0.7147	0.7712	0.6820	0.7663	0.7416	0.7361

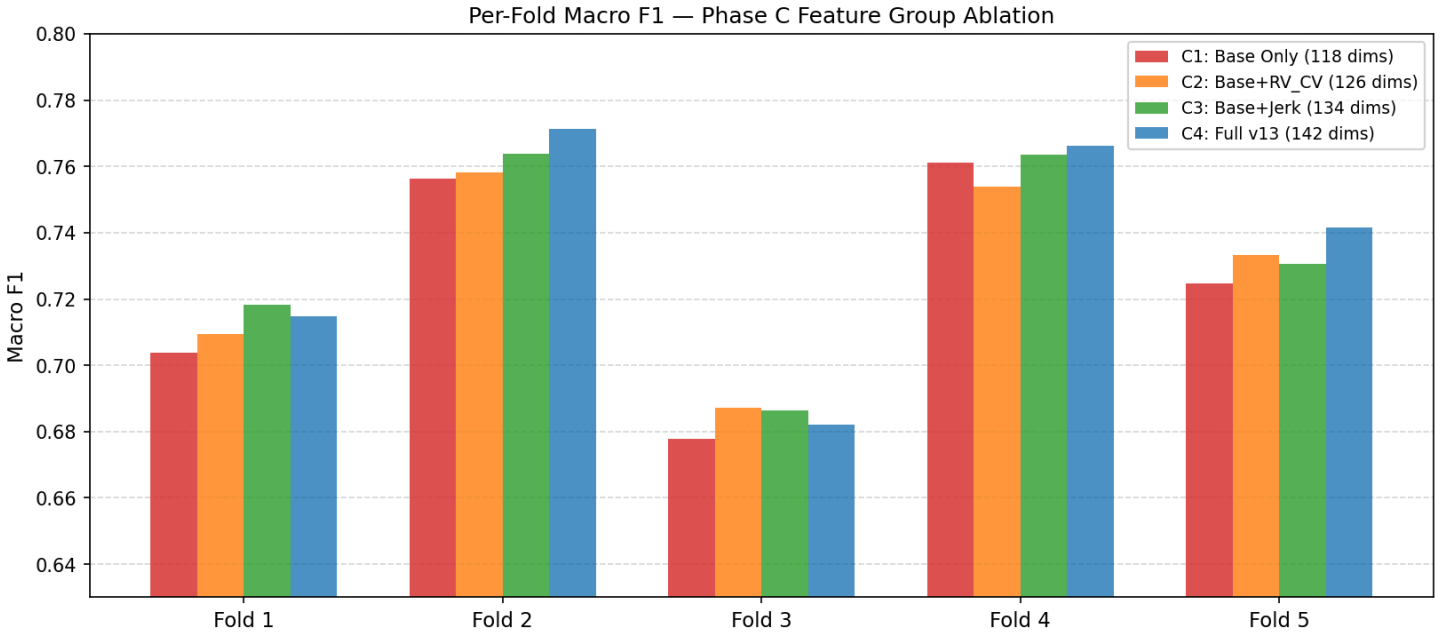
> *Source:* `ablation_phase_c.py` *— Phase C Feature Group Ablation, 4 Configurations × 5 Folds. Total wall-clock:* ***28.6 min****.*

Figure 4.1 — Mean Macro F1 by Configuration (bars = mean ± std; OOF F1 annotated)



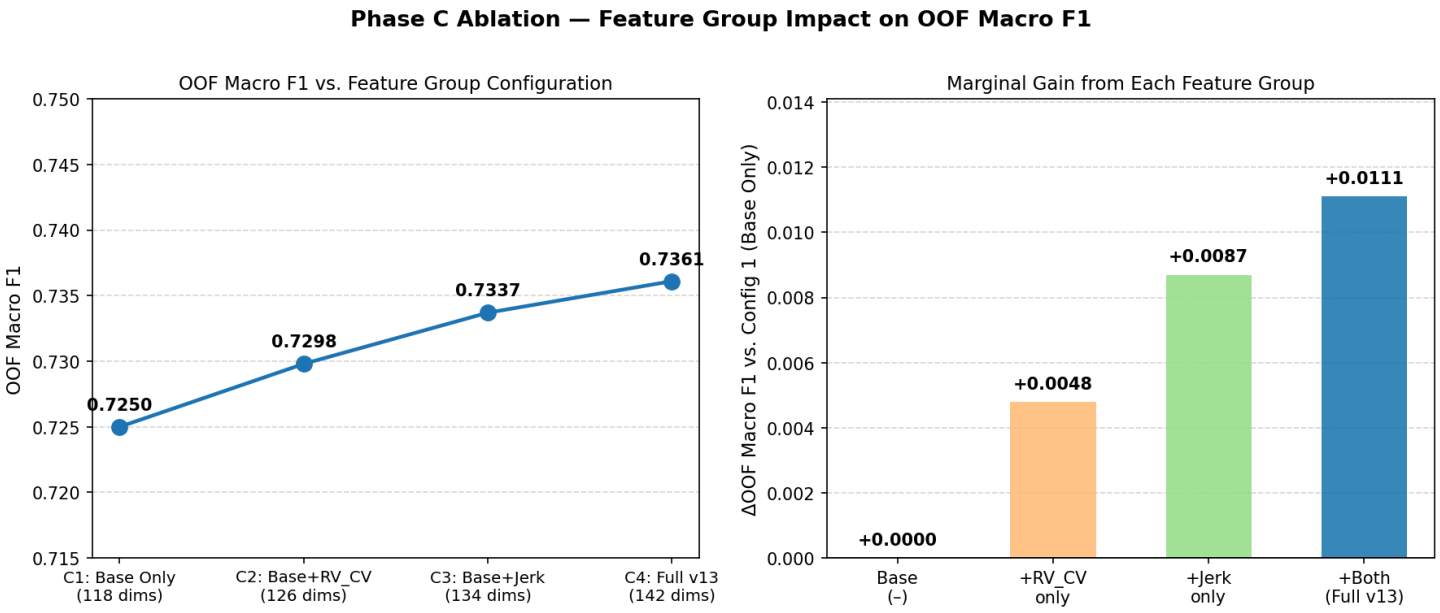
Monotonic improvement in both mean and OOF Macro F1 as feature groups are added. The gap between Config 1 (Base Only, OOF=0.7250) and Config 4 (Full v13, OOF=0.7361) represents the total contribution of the jerk and rolling_var_cv feature groups (+0.0111 OOF F1). Config 3 (Base+Jerk) gains more than Config 2 (Base+RV_CV), confirming jerk as the stronger individual contributor.

Figure 4.2 — Per-Fold Macro F1 Breakdown



Fold-by-fold comparison across all four configurations. The ranking Config 4 ≥ Config 3 ≥ Config 2 ≥ Config 1 holds consistently in Folds 2, 4, and 5. Fold 3 shows more variance across configs, likely due to a different subject-group composition placing more ambiguous L1/L2 samples in the validation set.

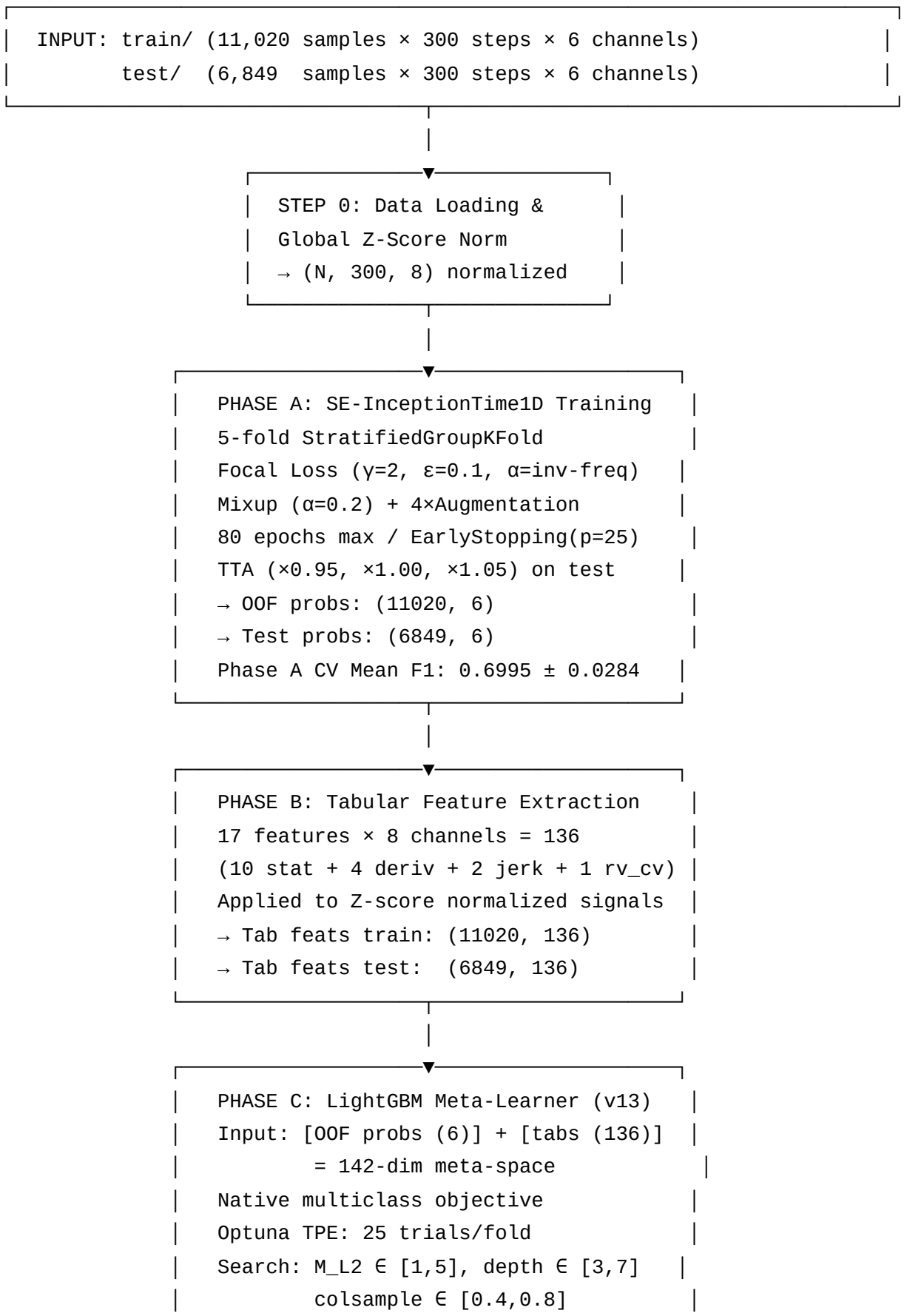
Figure 4.3 — Marginal Contribution: OOF F1 Trajectory and Delta vs. Base

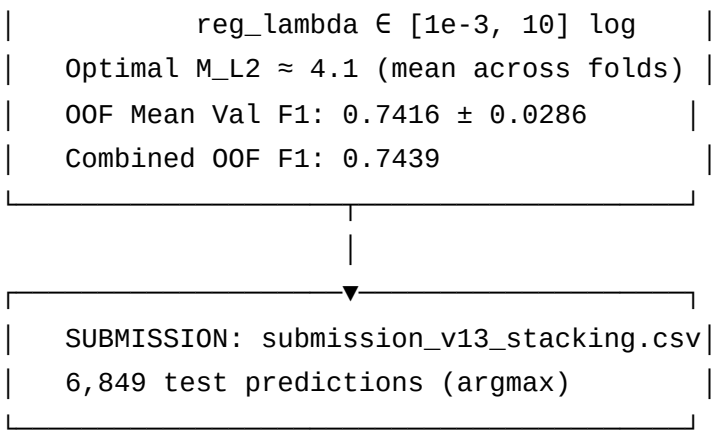


Left panel: OOF Macro F1 plotted as features are progressively added — the trajectory is monotonically increasing. Right panel: marginal gain relative to Config 1 baseline. Adding both jerk and RV_CV together (+0.0111) exceeds the sum of their individual contributions (+0.0087 + +0.0048 = +0.0135 if purely additive), indicating slight negative interaction — the two features share some redundancy via their proxy correlation with `deriv_zcr` and `std_mag`.

Conclusion: Both feature groups contribute positively and measurably to Phase C performance. Removing both `jerk` and `rolling_var_cv` simultaneously incurs the largest penalty (OOF F1: 0.7250 vs. 0.7361, **−0.0104**). `Jerk` is the stronger individual contributor: removing jerk alone costs −0.0067 OOF F1 (Config 2 vs. Config 4), while removing only `rolling_var_cv` costs −0.0026 (Config 3 vs. Config 4). Both feature groups provide genuine, additive discriminative value to the meta-learner and are retained in the v13 feature set.

Appendix: Pipeline Architecture Reference (v13)





Key Configuration Parameters (v13):

Parameter	Value
CNN: nb_filters	64 (hidden_dim = 256)
CNN: depth	6 SE-Inception blocks
CNN: bottleneck	32
CNN: SE reduction ratio	r = 16
CNN: dropout	0.4
Train: epochs (max)	80
Train: batch size	128
Train: learning rate	3×10^{-4} (AdamW)
Train: weight decay	1×10^{-3}
Train: LR schedule	CosineAnnealingLR (T_max=80, η_min=1×10^{-6})
Train: patience	25
Train: focal γ	2.0
Train: label smoothing	ε = 0.1
Train: Mixup α	0.2
Meta: Optuna trials/fold	25
Meta: M_L2 range	[1.0, 5.0]

Parameter	Value
Meta: LGB n_estimators	1000
Meta: LGB learning_rate	0.05
Meta: LGB subsample	0.8